

October 12, 2025

Martin Herzog

Stadtwerke München GmbH (SWM)
Emmy-Noether-Straße 2
80992 München

Subject: Application for Werkstudent:in – Asset Simulation Versorgungsnetze

Dear Martin Herzog,

SWM's vision of achieving **CO₂-neutral district heating by 2040** and operating Munich's energy networks with precision and sustainability deeply resonates with my own professional path in energy modeling and digital systems. As a Renewable Energy Engineer with hands-on experience developing simulation frameworks and analytical dashboards, I am eager to contribute to SWM's data-driven efforts in **asset simulation, scenario analysis, and network optimization**.

- **Asset & Network Simulation** Developed *PacCEM*, a **PyPSA-based capacity expansion model**, enabling optimization of renewable and storage assets under various constraints—skills directly applicable to SWM's asset-condition and renewal strategy simulations.
- **Data Processing & Automation** Built *EnergySim*, a unified GUI integrating OEMOF, PVLib, and WindPowerLib, automating large-scale simulations and generating ready-to-use datasets—reflecting SWM's focus on **data preparation, automation, and analysis of large data volumes**.
- **Heat Network Analysis** Modeled a **district-heating loop using TESPpy and MATLAB**, analyzing temperature and efficiency profiles to identify thermal losses and component optimization—aligned with SWM's **Fernwärme-Vision 2040** and geothermal heat initiatives.
- **Visualization & Decision Support** Created the *PyPSA Results Visualization Dashboard*, transforming raw simulation data into KPIs and time-series insights, ensuring transparency and interpretability—matching SWM's emphasis on **simulation-based asset evaluation** and reporting clarity.
- **Machine Learning & Cloud Readiness** Equipped with **Microsoft Azure AI Fundamentals** and certified in **IBM Machine Learning with Python**, I bring readiness to deploy predictive models for asset health assessment using cloud-based tools such as **Azure ML Studio**.

I am confident my technical expertise, from Python-based modeling to intuitive data visualization, can add tangible value to SWM's simulation and asset-management workflows. I am available for an interview at your earliest convenience and look forward to discussing how my work can support Munich's transition to a resilient, climate-neutral energy infrastructure.

Thank you for your time and consideration.

Sincerely,
Vaijayanth Sheri